

## **Erectile Dysfunction**

Patient Group Direction, version 008, issued 11 September 2026. Sildenafil and tadalafil, adult males aged 18 and over.

### **The two things that changed, and why**

- **THE NITRATE EXCLUSION NOW NAMES NICORANDIL AND AMYL NITRITE, IN BOTH ARMS.** The tadalafil arm said just "organic nitrates in any form", so poppers were invisible in the arm most likely to be requested by the men most likely to use them. Nicorandil appeared in neither arm at all.
- **THE CARDIOVASCULAR FITNESS CRITERION NOW HAS CONTENT.** Appendix 1 sets out the question to ask and what to do with the answer.
- **The alpha-blocker starting dose and the over-65 dose reduction, which the sildenafil arm had and the tadalafil arm had lost, now appear in both.**
- **These are not paperwork changes. A PDE5 inhibitor with a nitric oxide donor can kill, and the patient will not volunteer that he uses poppers unless he is asked directly.**

**Sildenafil**

## **Summary of BSSM and SmPC guidance for sildenafil and tadalafil in erectile dysfunction in adult men**

British Society for Sexual Medicine Guidelines on the Management of Erectile Dysfunction in Men 2017, Hackett G, Kirby M, Wylie K, Heald A, Ossei-Gerning N, Edwards D, Muneer A, Journal of Sexual Medicine 2018 (bssm.org.uk). Sildenafil Mylan 50 mg film-coated tablets, Summary of Product Characteristics, Viatris (PL 55183/0004), text revised May 2022, emc updated 28 July 2022. Cialis 10 mg film-coated tablets (tadalafil), Summary of Product Characteristics, Eli Lilly and Company Limited (PLGB 14895/0237), text revised 27 January 2026, emc updated 24 February 2026. Summarised 10 September 2026.

This summary is part 2 of the house format for a Get Real Health PGD: what the PGD is for, then the guidance that governs the condition, then the PGD itself. It is here so that a pharmacist can hold this document against the guidance it claims to follow. It summarises the source named above and adds nothing to it.

### **Erectile dysfunction as a marker of cardiovascular disease**

BSSM defines erectile dysfunction as the persistent inability to attain and/or maintain an erection sufficient for sexual performance.

BSSM states that the risk factors for erectile dysfunction (age, sedentary lifestyle, obesity, smoking, dyslipidaemia and the metabolic syndrome) are very similar to the established risk factors for cardiovascular disease, and that erectile dysfunction itself is a cardiovascular risk factor conferring a risk equivalent to a current moderate level of smoking.

BSSM describes erectile dysfunction as a sentinel marker for future cardiovascular events, occurring 3 to 5 years before an event, because the penile arteries are significantly smaller than the main coronary arteries.

BSSM states that a man with erectile dysfunction and no cardiac symptoms is a cardiac patient until proven otherwise, and that all men with unexplained erectile dysfunction should have a thorough evaluation with any coronary heart disease risk factors addressed.

BSSM says that measurement of fasting glucose and/or glycated haemoglobin, lipid profile and morning total testosterone should be considered mandatory in all newly presenting patients, because erectile dysfunction may be the first symptom of diabetes in up to 20% and hypogonadism is a treatable cause that may make men non-responsive to PDE5 inhibitors.

BSSM recommends a comprehensive medical, sexual and relationship history, a focused physical examination in all cases including blood pressure, heart rate, waist circumference and weight, and a genital examination.

Both SmPCs state that a medical history and physical examination should be undertaken to diagnose erectile dysfunction and determine potential underlying causes before pharmacological treatment is considered.

### **BSSM cardiovascular risk stratification**

BSSM states that the cardiac risk of sexual activity in properly assessed and advised patients with cardiovascular disease is minimal, and that sexual activity is no more stressful to the heart than many common daily activities.

BSSM's table of metabolic equivalents gives sexual intercourse with an established partner as 2 to 3 METs in the lower (normal) range and 5 to 6 METs for vigorous activity, comparable with walking 1 mile in 20 minutes on the level (3 to 4 METs), lifting and carrying 9 to 20 kg (4 to 5 METs), golf (4 to 5 METs), digging in the garden (3 to 5 METs) and heavy housework such as making beds or scrubbing floors (3 to 6 METs).

Low risk in the BSSM table comprises controlled hypertension, asymptomatic men with 3 or fewer risk factors for coronary artery disease (excluding age and gender), mild valvular disease, minimal or mild stable angina, post-successful revascularisation and heart failure NYHA class I; these men should be managed within primary care with treatment options reviewed with the patient and, where possible, his partner.

Intermediate risk comprises recent myocardial infarction or stroke within the last 6 weeks, asymptomatic men with more than 3 risk factors for coronary artery disease, left ventricular dysfunction or heart failure class II, a murmur of unknown cause, moderate stable angina, heart transplant and recurrent transient ischaemic attacks; BSSM recommends specialised evaluation (for example exercise testing for angina or echocardiography for a murmur) and re-assignment to the low- or high-risk group depending on the outcome.

High risk comprises severe, unstable or refractory angina, uncontrolled hypertension (systolic blood pressure above 180 mmHg), heart failure class III or IV, myocardial infarction or stroke within the last 14 days, high-risk arrhythmias, hypertrophic cardiomyopathy and moderate or severe valve disease; BSSM says these men should be referred for specialised cardiac evaluation and treatment for erectile dysfunction deferred until the cardiac condition is established or specialist evaluation completed.

BSSM states that patients who remain in the high-risk group should not be offered treatment for erectile dysfunction in primary care, and that their management should be supervised by a specialist team that will probably include a cardiologist.

BSSM adds that there is no evidence that currently licensed treatments for erectile dysfunction add to the overall cardiovascular risk in patients with or without previously diagnosed cardiovascular disease.

### **Red flags and referral**

BSSM lists indications for specialist investigation as young patients who have always had difficulty obtaining or sustaining an erection, patients with a history of trauma, an abnormality of the testicles or penis found on examination, and patients unresponsive to medical therapies who may want surgical treatment.

BSSM says a genital examination is essential where there is a history of rapid onset of pain, deviation of the penis during tumescence, symptoms of hypogonadism or other urological symptoms, and describes Peyronie's disease (a lump in the shaft, pain in the shaft, abnormal angulation of the erect penis).

BSSM recommends that if serum testosterone is borderline or low it should be repeated on a further morning sample with luteinising hormone and prolactin, with discussion with or referral to a specialist clinic if results are abnormal, and that patients with persistent unexplained hyperprolactinaemia should be referred to an endocrinologist.

BSSM states that about 25 to 50% of patients do not respond to PDE5 inhibitors within 12 months, that patients should receive at least 4 and preferably 8 doses of the highest tolerated dose of at least 2 drugs (sequentially, not concurrently) with

adequate sexual stimulation before being classed as non-responders, and that inadequate prescribing or instruction is the major cause of failure.

BSSM recommends follow-up ideally within 6 weeks of commencing therapy.

Both SmPCs state that an erection persisting longer than 4 hours requires immediate medical assistance, because untreated priapism can cause penile tissue damage and permanent loss of potency.

Both SmPCs advise that in the event of any sudden visual defect the patient should stop the medicine and consult a physician immediately; the tadalafil SmPC adds the same instruction for sudden decrease or loss of hearing.

#### **PDE5 inhibitor doses, timing, titration and adjustments (SmPCs)**

The recommended sildenafil dose is 50 mg taken as needed approximately one hour before sexual activity; based on efficacy and tolerability the dose may be increased to 100 mg or decreased to 25 mg.

The maximum recommended sildenafil dose is 100 mg and the maximum recommended dosing frequency is once per day. Sexual stimulation is required for sildenafil to be effective, and if taken with food the onset of activity may be delayed compared with the fasted state.

Sildenafil dosage adjustments are not required in elderly patients aged 65 and over, and the standard dosing applies in mild to moderate renal impairment (creatinine clearance 30 to 80 ml/min).

In severe renal impairment (creatinine clearance below 30 ml/min) and in hepatic impairment such as cirrhosis, a sildenafil dose of 25 mg should be considered, with step-wise increase to 50 mg and up to 100 mg as necessary.

A sildenafil starting dose of 25 mg should be considered in patients receiving CYP3A4 inhibitors and in patients on alpha-blocker therapy, who should be stabilised on the alpha-blocker before sildenafil is started.

The recommended tadalafil dose is 10 mg taken prior to anticipated sexual activity, with or without food; in patients in whom 10 mg does not produce an adequate effect, 20 mg might be tried.

Tadalafil may be taken at least 30 minutes before sexual activity, the maximum dose frequency is once per day, and 10 mg and 20 mg doses are not recommended for continuous daily use.

For patients who anticipate frequent use (at least twice weekly), a once-daily regimen of 5 mg (reducible to 2.5 mg) may be considered, with the appropriateness of continued daily use reassessed periodically.

Dose adjustments are not required in elderly men, in mild to moderate renal impairment, or in diabetes.

In severe renal impairment 10 mg is the maximum recommended dose and once-daily dosing is not recommended.

In hepatic impairment the recommended dose is 10 mg; there are no data on doses above 10 mg in hepatic impairment, and in severe hepatic impairment (Child-Pugh C) clinical data are limited so a careful individual benefit-risk evaluation is required.

The SmPC states it is not known whether tadalafil is effective in patients who have undergone pelvic surgery or radical non-nerve-sparing prostatectomy.

#### **Absolute contraindications**

BSSM states that absolute contraindications to PDE5 inhibitors are regular or intermittent use of nitrates in any form, including amyl nitrite (poppers), nicorandil, and guanylate cyclase stimulators such as riociguat, because combined use could cause unpredictable falls in blood pressure and potentially catastrophic hypotension.

Both SmPCs contraindicate co-administration with nitrates or nitric oxide donors in any form, and with riociguat.

Both SmPCs contraindicate use in men for whom sexual activity is inadvisable, for example patients with severe cardiovascular disorders such as unstable angina or severe cardiac failure.

Both SmPCs contraindicate use in patients who have loss of vision in one eye because of non-arteritic anterior ischaemic optic neuropathy (NAION), regardless of whether the episode was connected with previous PDE5 inhibitor exposure.

The sildenafil SmPC contraindicates use in severe hepatic impairment, hypotension (blood pressure below 90/50 mmHg), a recent history of stroke or myocardial infarction, and known hereditary degenerative retinal disorders such as retinitis pigmentosa, because safety has not been studied in these groups.

The tadalafil SmPC contraindicates use in patients with myocardial infarction within the last 90 days, unstable angina or angina during sexual intercourse, NYHA class 2 or greater heart failure in the last 6 months, uncontrolled arrhythmias, hypotension (below 90/50 mmHg) or uncontrolled hypertension, and stroke within the last 6 months.

The tadalafil SmPC does not list severe hepatic impairment as a contraindication but states that clinical data are limited and a careful individual benefit-risk evaluation should be undertaken.

#### **Cautions**

Both SmPCs state that prescribers should consider the patient's cardiovascular status before initiating treatment, because there is a degree of cardiac risk associated with sexual activity and PDE5 inhibitors have vasodilator properties causing mild, transient decreases in blood pressure.

The sildenafil SmPC identifies patients with increased susceptibility to vasodilators, including those with left ventricular outflow obstruction (aortic stenosis, hypertrophic obstructive cardiomyopathy) and multiple system atrophy with severely impaired autonomic control of blood pressure.

Both SmPCs advise caution in patients with anatomical deformation of the penis (angulation, cavernosal fibrosis or Peyronie's disease) and in conditions predisposing to priapism (sickle cell anaemia, multiple myeloma or leukaemia).

Both SmPCs state that combination with other PDE5 inhibitors or other treatments for erectile dysfunction has not been studied and is not recommended.

The sildenafil SmPC advises use only after careful benefit-risk assessment in patients with bleeding disorders or active peptic ulceration, and states that serious cardiovascular events have been reported post-marketing in temporal association with sildenafil, mostly in patients with pre-existing cardiovascular risk factors.

The tadalafil SmPC reports visual defects including central serous chorioretinopathy and NAION, and cases of sudden hearing loss, and advises patients to stop and seek prompt medical attention if these occur.

BSSM notes that risk factors for NAION (age over 50, heart disease, diabetes, hypertension, dyslipidaemia, nicotine use) are the same factors that make men likely to have erectile dysfunction.

#### **Interactions with alpha-blockers and CYP3A4 inhibitors**

BSSM states that alpha-blockers may interact with PDE5 inhibitors to cause orthostatic hypotension, that sildenafil should be initiated at the lowest dose only if the patient is stabilised on alpha-blocker therapy with dosing separated by at least 4 hours, and that these interactions are rarely clinically significant when the drugs are not started simultaneously.

The sildenafil SmPC says symptomatic hypotension with alpha-blockers is most likely within 4 hours of dosing, that patients should be haemodynamically stable on the alpha-blocker first, that a 25 mg starting dose should be considered, and that patients should be told what to do if postural hypotensive symptoms occur.

The tadalafil SmPC states that the combination of tadalafil and doxazosin is not recommended because it significantly increases the blood pressure-lowering effect for at least twelve hours and may cause syncope; with other alpha-blockers, including alfuzosin and tamsulosin, caution is advised, notably in the elderly, with treatment started at minimal dosage.

BSSM says dose adjustment of all PDE5 inhibitors may be needed in patients over 65, in hepatic impairment, in renal impairment, and with potent CYP3A4 inhibitors such as ritonavir, erythromycin and cimetidine.

The sildenafil SmPC states that CYP3A4 inhibitors (ketoconazole, erythromycin, cimetidine) reduce sildenafil clearance and a 25 mg starting dose should be considered; co-administration with ritonavir is not advised and in any event the dose should not exceed 25 mg within 48 hours.

The tadalafil SmPC advises caution with potent CYP3A4 inhibitors (ritonavir, saquinavir, ketoconazole, itraconazole and erythromycin), and lists clarithromycin and grapefruit juice as other inhibitors expected to increase tadalafil exposure.

Both SmPCs note that CYP3A4 inducers such as rifampicin reduce PDE5 inhibitor exposure; the tadalafil SmPC also names phenobarbital, phenytoin and carbamazepine.

The sildenafil SmPC advises caution when sildenafil is initiated in patients treated with sacubitril/valsartan, and states that nicorandil has a nitrate component with the potential for a serious interaction.

BSSM states that co-administration with antihypertensives produces a small additive drop in blood pressure that does not usually cause significant orthostatic hypotension (8.4/5.5 mmHg for sildenafil, 1.6/0.8 mmHg for tadalafil).

BSSM and the tadalafil SmPC state that if a patient taking a PDE5 inhibitor develops chest pain, nitroglycerine must be withheld for at least 24 hours after sildenafil and at least 48 hours after tadalafil, and then given only under close medical observation.

#### **Adverse effects and what to tell the patient**

The sildenafil SmPC lists headache as very common and dizziness, visual colour distortion, visual disturbance, blurred vision, flushing, hot flush, nasal congestion, nausea and dyspepsia as common.

The tadalafil SmPC lists headache, flushing, nasal congestion, dyspepsia, back pain, myalgia and pain in extremity as common, with incidence increasing with dose; adverse reactions are described as transient and generally mild or moderate.

Both SmPCs state that patients should be aware of how they react to the medicine before driving or operating machinery, since dizziness and altered vision have been reported.

BSSM states that PDE5 inhibitors are not initiators of erection and require sexual stimulation, and that so-called failure is often due to sub-optimal counselling at the first consultation, so the patient should understand how to take the tablets properly and return if dissatisfied.

BSSM states that food, particularly fatty food, interacts most with sildenafil and least with tadalafil, and that no interaction with alcohol has been observed with any PDE5 inhibitor up to moderate concentrations; the tadalafil SmPC notes postural dizziness and orthostatic hypotension in some subjects with larger amounts of alcohol.

BSSM recommends that lifestyle change and risk-factor management (exercise, smoking cessation, weight and lipid control) should accompany all treatment regimens, but that pharmacotherapy should not be withheld because lifestyle changes have not been made.

BSSM states that once sildenafil came off patent it was removed from Schedule 2 so that it can be prescribed irrespective of comorbidities and without restriction on frequency, and that the SmPC allows up to 100 mg in any 24-hour period prior to anticipated sexual activity.

#### **What the sources do not say**

The BSSM guideline's risk stratification figure is presented as an image in the PDF; the categories above are taken from the accompanying table text in the guideline.

The BSSM guideline gives the 6-week and 14-day windows for recent myocardial infarction or stroke in its risk table but does not reconcile these with the SmPC contraindication periods (sildenafil: recent history, undefined; tadalafil: 90 days for myocardial infarction and 6 months for stroke).

Neither SmPC gives a minimum interval between a PDE5 inhibitor dose and an alpha-blocker dose; the 4-hour separation for sildenafil comes from BSSM.

The BSSM guideline does not give tadalafil-specific alpha-blocker advice; that comes from the Cialis SmPC.

Neither source sets a specific upper age limit for treatment or a maximum quantity to supply per consultation.

**Patient Group Direction for the supply of sildenafil 25mg, 50mg and 100mg film-coated tablets for the treatment of erectile dysfunction in adult males aged 18 years and over.**

## Healthcare professionals covered, and training

- Pharmacist registered and practising with the GPhC.
- Pharmacy technician registered and practising with the GPhC.
- All users must be familiar with the current SPC for the product supplied, the BNF and NICE CKS on erectile dysfunction.
- ALL USERS MUST BE ABLE TO RECOGNISE EVERY NITRIC OXIDE DONOR BY NAME, not only the medicines with "nitrate" in the title. That includes nicorandil, which is an angina medicine, and amyl nitrite or "poppers", which is bought recreationally and will not appear on any medication list.
- All users must be able to apply the cardiovascular fitness assessment in Appendix 1 and to record its result.
- All users must understand that erectile dysfunction can be the first sign of cardiovascular disease or of undiagnosed diabetes, and must be willing to refer rather than treat where the history suggests it.
- All users must previously have supplied medicines under a PGD.
- All users must understand capacity and consent and be aware of the Mental Capacity Act 2005.
- All users must be up to date with CPD and appraisal, and with MHRA safety alerts relating to these medicines.
- All users must hold appropriate indemnity covering this service, must work in compliance with the SOPs of their own employer, and must practise only within the bounds of their own competence.

## The PGD

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| <b>Indication</b>                       | Treatment of erectile dysfunction in adult males aged 18 years and over, being the inability to achieve or maintain a penile erection sufficient for satisfactory sexual performance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Inclusion criteria</b>               | <ul style="list-style-type: none"> <li>• Adult male aged 18 years and over.</li> <li>• Clinical presentation consistent with erectile dysfunction.</li> <li>• <b>Cardiovascular fitness assessment in Appendix 1 completed, passed, and recorded.</b></li> <li>• <b>Blood pressure measured today and within the acceptable range.</b></li> <li>• <b>Nitrate and nitric oxide donor questions asked, including the direct question about poppers, and all answers negative.</b></li> <li>• Valid informed consent obtained.</li> <li>• No exclusion criterion present.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Nitrates and nitric oxide donors</b> | <p><b>ABSOLUTE EXCLUSION IN BOTH ARMS. The combination causes profound, prolonged and potentially fatal hypotension. Ask about every item on this list by name.</b></p> <ul style="list-style-type: none"> <li>• <b>Glyceryl trinitrate, in any form: tablets, spray, patches, ointment, or an infusion. Including a GTN spray carried "just in case" and used rarely.</b></li> <li>• Isosorbide mononitrate or isosorbide dinitrate.</li> <li>• <b>NICORANDIL. This is a potassium channel activator with a nitrate moiety and it is a nitric oxide donor. It is prescribed for angina and does not have "nitrate" in its name, which is why it gets missed.</b></li> <li>• <b>AMYL NITRITE, butyl nitrite, isobutyl nitrite or alkyl nitrites, commonly sold as "poppers". These are bought recreationally, are not prescribed, and will not appear on any medication list. YOU MUST ASK DIRECTLY.</b></li> <li>• Sodium nitroprusside.</li> <li>• Riociguat, and any other soluble guanylate cyclase stimulator.</li> </ul> <p><b>How to ask about poppers, so that the answer is useful: "Some men use poppers, amyl nitrite, at the same time as these tablets. Taken together they can drop your blood pressure to a dangerous level. Do you ever use them, or think you might?" Ask it without judgement and record the answer.</b></p> <p><b>Where the patient uses poppers and is unwilling to stop, do not supply. Explain why plainly: it is the combination that is dangerous, not either</b></p> |

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|                        | one alone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Cardiovascular fitness | <p>Apply the following and record the result.</p> <p><b>THE FUNCTIONAL TEST.</b> Ask: "Can you walk a mile on the flat in about 20 minutes, or climb two flights of stairs briskly, without chest pain and without having to stop for breath?"</p> <ul style="list-style-type: none"> <li>• YES, comfortably: proceed. Record the answer given.</li> <li>• NO, or the patient does not know because he never exerts himself that much: <b>DO NOT SUPPLY.</b> Refer to the GP for cardiovascular assessment. Sexual activity carries a similar cardiac workload.</li> <li>• Chest pain, breathlessness or palpitations on exertion OR during previous sexual activity: <b>DO NOT SUPPLY.</b> Refer.</li> </ul> <p>Also record, at every supply:</p> <ul style="list-style-type: none"> <li>• Blood pressure measured today. Do not accept a figure from memory.</li> <li>• Whether the patient has known cardiovascular disease, and if so what.</li> <li>• Date of the last cardiovascular review, where known.</li> </ul> <p><b>Erectile dysfunction commonly precedes a cardiovascular event by two to five years and is frequently the first presentation of undiagnosed diabetes. A man attending for this is an opportunity, not just a customer. Where he has not had a recent check, recommend one and record that you did.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Exclusion criteria     | <p><b>Refer, do not supply, where any of the following applies.</b></p> <ul style="list-style-type: none"> <li>• Known hypersensitivity to sildenafil or to any excipient.</li> <li>• Under 18 years of age. Female.</li> <li>• <b>ANY nitrate or nitric oxide donor, including nicorandil and amyl nitrite. See the row above.</b></li> <li>• Riociguat or any other soluble guanylate cyclase stimulator.</li> <li>• Recent stroke or myocardial infarction, within the last 6 months.</li> <li>• Hypotension, blood pressure below 90/50 mmHg, measured today.</li> <li>• Uncontrolled hypertension, blood pressure above 170/100 mmHg, measured today.</li> <li>• <b>Fails the cardiovascular fitness assessment in Appendix 1, or it has not been done.</b></li> <li>• Unstable angina, angina during sexual activity, heart failure of NYHA class 2 or greater in the last 6 months, uncontrolled arrhythmia, hypertrophic cardiomyopathy, significant aortic stenosis or other moderate to severe valve disease, or a murmur of unknown cause.</li> <li>• Known hereditary degenerative retinal disorder, for example retinitis pigmentosa.</li> <li>• Taking ritonavir or cobicistat. Refer (the sildenafil dose must not exceed 25mg in 48 hours and cannot be titrated under this PGD).</li> <li>• Previous episode of non-arteritic anterior ischaemic optic neuropathy.</li> <li>• <b>Previous priapism, or an erection lasting more than 4 hours on any previous PDE5 inhibitor.</b></li> <li>• <b>Already taking any other PDE5 inhibitor, including one obtained online or from another supplier. Do not add a second.</b></li> <li>• <b>Erectile dysfunction of sudden onset following trauma, surgery or a new medicine, or accompanied by penile pain or deformity. Refer for a diagnosis rather than treating the symptom.</b></li> <li>• Valid informed consent not obtained.</li> <li>• Severe hepatic impairment.</li> </ul> |
| Cautions               | <p>Anatomical deformation of the penis: angulation, cavernosal fibrosis or Peyronie's disease.</p> <p>Conditions predisposing to priapism: sickle cell anaemia, multiple myeloma,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|                                                       | <p>leukaemia.</p> <p>Active peptic ulceration or a bleeding disorder.</p> <p><b>Advise the patient that sudden visual loss may indicate NAION, and sudden hearing loss has also been reported. Stop the medicine and seek urgent medical attention for either.</b></p> <p><b>These medicines do not protect against sexually transmitted infections. Say so, and offer testing where appropriate.</b></p> <p><b>Sexual stimulation is required. The tablet does not produce an erection by itself. Failure to explain this is the commonest reason a patient reports the medicine "does not work".</b></p> <p><b>ALPHA-BLOCKERS. The patient must be stable on his alpha-blocker before starting. START AT 25mg. Do not start at 50mg.</b></p> <p><b>AGED 65 OR OVER. Start at 25mg.</b></p> <p><b>CYP3A4 inhibitors, for example erythromycin, clarithromycin, ketoconazole or itraconazole. Start at 25mg. Ritonavir and cobicistat exclude.</b></p> <p>Mild to moderate hepatic impairment. Start at 25mg.</p> <p>Severe renal impairment, eGFR below 30 mL/min. Start at 25mg.</p> <p>Avoid grapefruit juice, which increases exposure.</p> |
| <b>Actions if the patient is excluded or declines</b> | <p>Explain why supply is not appropriate and what to do instead. Where the exclusion is cardiovascular, say so plainly and recommend a GP assessment: erectile dysfunction is frequently the first sign of cardiovascular disease, and a referral here is worth more to the patient than a tablet. Where the exclusion is poppers, explain that it is the combination that is dangerous. Document the advice given and the decision reached. Inform or refer to the GP as appropriate.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## The medicine

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|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name, form and strength</b>             | Sildenafil 25mg, 50mg and 100mg film-coated tablets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Legal category</b>                      | POM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Dose and frequency</b>                  | <ul style="list-style-type: none"> <li>Starting dose 50mg as needed, about 1 hour before sexual activity.</li> <li><b>START AT 25mg where the patient is aged 65 or over, on an alpha-blocker, on a CYP3A4 inhibitor, or has hepatic or severe renal impairment.</b></li> <li>May be increased to a maximum of 100mg, or decreased to 25mg, on efficacy and tolerability.</li> <li><b>MAXIMUM ONE DOSE IN ANY 24 HOURS.</b></li> <li>Efficacy may be reduced after a high-fat meal.</li> <li>Can be taken from 30 minutes to 4 hours before sexual activity. Sexual stimulation is required.</li> </ul> |
| <b>Quantity to be supplied</b>             | Up to 8 tablets per supply, being about one month at twice-weekly use. Decide the quantity on individual need and record the decision. Do not supply more than 8.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Maximum or minimum treatment period</b> | No maximum treatment period. Review effectiveness, blood pressure and cardiovascular fitness at least annually, and reassess where there has been no improvement after 6 to 8 attempts at the maximum tolerated dose.                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Route and method</b>                    | Oral.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Storage</b>                             | Store below 30 degrees Celsius in the original packaging.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Adverse effects</b>                     | Very common: headache. Common: flushing, dyspepsia, nasal congestion, dizziness, visual disturbance including a colour tinge, increased brightness or blurred vision, nausea. Uncommon: vomiting, rash, epistaxis, palpitations, tachycardia, drowsiness, eye pain, tinnitus. Rare but serious: priapism, non-                                                                                                                                                                                                                                                                                          |

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|                              | arteritic anterior ischaemic optic neuropathy, sudden sensorineural hearing loss.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Yellow Card reporting</b> | Report suspected adverse reactions via <a href="https://yellowcard.mhra.gov.uk">https://yellowcard.mhra.gov.uk</a> and inform the GP as appropriate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Records to be kept</b>    | <ul style="list-style-type: none"> <li>• That valid informed consent was given.</li> <li>• Patient name, address, date of birth, and the GP with whom they are registered.</li> <li>• <b>THE ANSWER TO THE NITRATE AND NITRIC OXIDE DONOR QUESTIONS, including the direct question about poppers.</b></li> <li>• <b>THE CARDIOVASCULAR FITNESS ANSWER, in the patient's terms, and the blood pressure measured today.</b></li> <li>• <b>Whether the patient is on an alpha-blocker, and if so which one and whether he is stable on it.</b></li> <li>• Age, and where aged 65 or over, the starting dose chosen.</li> <li>• Any CYP3A4 inhibitor, and any hepatic or renal impairment, and the starting dose chosen as a result.</li> <li>• Name of the healthcare practitioner and registration number.</li> <li>• Name, brand, form, strength, dose, route, quantity and date of supply.</li> <li>• That the patient was advised these medicines do not protect against sexually transmitted infections.</li> <li>• Advice given, including advice given if the patient is excluded or declines.</li> <li>• Details of any adverse drug reactions and the actions taken.</li> <li>• That the medicine was supplied under this PGD.</li> </ul> <p>Records signed, dated, legible and contemporaneous. Electronic records are acceptable. Keep for 8 years.</p> |
| <b>Disposal</b>              | Advise the patient to return unused tablets to a pharmacy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Information for the patient

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|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Written information</b>           | Supply the patient information leaflet provided with the product.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Counselling</b>                   | <ul style="list-style-type: none"> <li>• Take about an hour before sex. It can work from 30 minutes up to about 4 hours.</li> <li>• <b>You still need to be aroused. The tablet does not do it on its own.</b></li> <li>• <b>One tablet in 24 hours. No more.</b></li> <li>• A heavy, fatty meal can make it work less well.</li> <li>• Avoid grapefruit juice.</li> <li>• <b>NEVER take this with poppers, or with any nitrate medicine such as a GTN spray or nicorandil. Together they can drop your blood pressure to a dangerous level.</b></li> <li>• This does not protect you against sexually transmitted infections.</li> </ul> |
| <b>Follow-up and urgent symptoms</b> | <p><b>Seek immediate medical attention for:</b></p> <ul style="list-style-type: none"> <li>• <b>An erection lasting more than 4 hours. Go to A&amp;E. This can cause permanent damage.</b></li> <li>• <b>Sudden loss of vision in one or both eyes. Stop the tablets.</b></li> <li>• Sudden loss or reduction of hearing.</li> <li>• <b>Chest pain during or after sex. Do not take a GTN spray for it; tell the paramedics you have taken this medicine.</b></li> </ul> <p>If you are not under regular GP review, arrange a cardiovascular and diabetes check. Erection problems are often the first sign of something else.</p>        |

## Tadalafil

**Patient Group Direction for the supply of tadalafil 2.5mg, 5mg, 10mg and 20mg film-coated tablets for the treatment of erectile dysfunction in adult males aged 18 years and over, on demand or once daily.**

### Healthcare professionals covered, and training

- Pharmacist registered and practising with the GPhC.
- Pharmacy technician registered and practising with the GPhC.
- All users must be familiar with the current SPC for the product supplied, the BNF and NICE CKS on erectile dysfunction.
- ALL USERS MUST BE ABLE TO RECOGNISE EVERY NITRIC OXIDE DONOR BY NAME, not only the medicines with "nitrate" in the title. That includes nicorandil, which is an angina medicine, and amyl nitrite or "poppers", which is bought recreationally and will not appear on any medication list.
- All users must be able to apply the cardiovascular fitness assessment in Appendix 1 and to record its result.
- All users must understand that erectile dysfunction can be the first sign of cardiovascular disease or of undiagnosed diabetes, and must be willing to refer rather than treat where the history suggests it.
- All users must previously have supplied medicines under a PGD.
- All users must understand capacity and consent and be aware of the Mental Capacity Act 2005.
- All users must be up to date with CPD and appraisal, and with MHRA safety alerts relating to these medicines.
- All users must hold appropriate indemnity covering this service, must work in compliance with the SOPs of their own employer, and must practise only within the bounds of their own competence.

### The PGD

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indication                       | Treatment of erectile dysfunction in adult males aged 18 years and over. Available as on-demand dosing, or once-daily dosing for men who anticipate frequent sexual activity, at least twice per week.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Inclusion criteria               | <ul style="list-style-type: none"><li>• Adult male aged 18 years and over.</li><li>• Clinical presentation consistent with erectile dysfunction.</li><li>• <b>Cardiovascular fitness assessment in Appendix 1 completed, passed, and recorded.</b></li><li>• <b>Blood pressure measured today and within the acceptable range.</b></li><li>• <b>Nitrate and nitric oxide donor questions asked, including the direct question about poppers, and all answers negative.</b></li><li>• Valid informed consent obtained.</li><li>• No exclusion criterion present.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Nitrates and nitric oxide donors | <p><b>ABSOLUTE EXCLUSION IN BOTH ARMS. The combination causes profound, prolonged and potentially fatal hypotension. Ask about every item on this list by name.</b></p> <ul style="list-style-type: none"><li>• <b>Glyceryl trinitrate, in any form: tablets, spray, patches, ointment, or an infusion. Including a GTN spray carried "just in case" and used rarely.</b></li><li>• Isosorbide mononitrate or isosorbide dinitrate.</li><li>• <b>NICORANDIL. This is a potassium channel activator with a nitrate moiety and it is a nitric oxide donor. It is prescribed for angina and does not have "nitrate" in its name, which is why it gets missed.</b></li><li>• <b>AMYL NITRITE, butyl nitrite, isobutyl nitrite or alkyl nitrites, commonly sold as "poppers". These are bought recreationally, are not prescribed, and will not appear on any medication list. YOU MUST ASK DIRECTLY.</b></li><li>• Sodium nitroprusside.</li><li>• Riociguat, and any other soluble guanylate cyclase stimulator.</li></ul> <p><b>How to ask about poppers, so that the answer is useful: "Some men use poppers, amyl nitrite, at the same time as these tablets. Taken together they can drop your blood pressure to a dangerous level. Do you ever use them, or think you might?" Ask it without judgement and record the answer.</b></p> |

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|                        | <p>Where the patient uses poppers and is unwilling to stop, do not supply. Explain why plainly: it is the combination that is dangerous, not either one alone.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Cardiovascular fitness | <p>Apply the following and record the result.</p> <p><b>THE FUNCTIONAL TEST.</b> Ask: "Can you walk a mile on the flat in about 20 minutes, or climb two flights of stairs briskly, without chest pain and without having to stop for breath?"</p> <ul style="list-style-type: none"> <li>• YES, comfortably: proceed. Record the answer given.</li> <li>• NO, or the patient does not know because he never exerts himself that much: <b>DO NOT SUPPLY.</b> Refer to the GP for cardiovascular assessment. Sexual activity carries a similar cardiac workload.</li> <li>• Chest pain, breathlessness or palpitations on exertion OR during previous sexual activity: <b>DO NOT SUPPLY.</b> Refer.</li> </ul> <p>Also record, at every supply:</p> <ul style="list-style-type: none"> <li>• Blood pressure measured today. Do not accept a figure from memory.</li> <li>• Whether the patient has known cardiovascular disease, and if so what.</li> <li>• Date of the last cardiovascular review, where known.</li> </ul> <p><b>Erectile dysfunction commonly precedes a cardiovascular event by two to five years and is frequently the first presentation of undiagnosed diabetes. A man attending for this is an opportunity, not just a customer. Where he has not had a recent check, recommend one and record that you did.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Exclusion criteria     | <p><b>Refer, do not supply, where any of the following applies.</b></p> <ul style="list-style-type: none"> <li>• Known hypersensitivity to tadalafil or to any excipient.</li> <li>• Under 18 years of age. Female.</li> <li>• <b>ANY nitrate or nitric oxide donor, including nicorandil and amyl nitrite. See the row above.</b></li> <li>• Riociguat or any other soluble guanylate cyclase stimulator.</li> <li>• Recent stroke or myocardial infarction, within the last 6 months.</li> <li>• Hypotension, blood pressure below 90/50 mmHg, measured today.</li> <li>• Uncontrolled hypertension, blood pressure above 170/100 mmHg, measured today.</li> <li>• <b>Fails the cardiovascular fitness assessment in Appendix 1, or it has not been done.</b></li> <li>• Unstable angina, angina during sexual activity, heart failure of NYHA class 2 or greater in the last 6 months, uncontrolled arrhythmia, hypertrophic cardiomyopathy, significant aortic stenosis or other moderate to severe valve disease, or a murmur of unknown cause.</li> <li>• Known hereditary degenerative retinal disorder, for example retinitis pigmentosa.</li> <li>• Taking doxazosin. The combination with tadalafil is not recommended in the SmPC; use the sildenafil arm or refer.</li> <li>• Previous episode of non-arteritic anterior ischaemic optic neuropathy.</li> <li>• <b>Previous priapism, or an erection lasting more than 4 hours on any previous PDE5 inhibitor.</b></li> <li>• <b>Already taking any other PDE5 inhibitor, including one obtained online or from another supplier. Do not add a second.</b></li> <li>• <b>Erectile dysfunction of sudden onset following trauma, surgery or a new medicine, or accompanied by penile pain or deformity. Refer for a diagnosis rather than treating the symptom.</b></li> <li>• Valid informed consent not obtained.</li> <li>• Severe hepatic impairment, Child-Pugh class C.</li> <li>• Severe renal impairment, creatinine clearance below 30 mL/min: once-daily dosing is excluded. On-demand dosing is permitted at a</li> </ul> |

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|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | maximum of 10mg (see cautions).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Cautions</b>                                       | <p>Anatomical deformation of the penis: angulation, cavernosal fibrosis or Peyronie's disease.</p> <p>Conditions predisposing to priapism: sickle cell anaemia, multiple myeloma, leukaemia.</p> <p>Active peptic ulceration or a bleeding disorder.</p> <p><b>Advise the patient that sudden visual loss may indicate NAION, and sudden hearing loss has also been reported. Stop the medicine and seek urgent medical attention for either.</b></p> <p><b>These medicines do not protect against sexually transmitted infections. Say so, and offer testing where appropriate.</b></p> <p><b>Sexual stimulation is required. The tablet does not produce an erection by itself. Failure to explain this is the commonest reason a patient reports the medicine "does not work".</b></p> <p><b>PROLONGED HALF-LIFE</b>, about 17.5 hours. Effects and adverse effects last considerably longer than with sildenafil, and so does the interaction with any nitrate. A patient who takes tadalafil on Friday is still at risk on Sunday. Say so.</p> <p><b>ALPHA-BLOCKERS.</b> The patient must be stable on his alpha-blocker before starting. Start on-demand dosing at 10mg and do not exceed it until tolerance is established. For once-daily dosing, start at 2.5mg.</p> <p><b>AGED 65 OR OVER.</b> Start on-demand dosing at 10mg rather than escalating, and once-daily dosing at 2.5mg.</p> <p><b>Potent CYP3A4 inhibitors, including ritonavir and cobicistat, and for example ketoconazole, itraconazole or clarithromycin. On-demand dose must not exceed 10mg in any 72 hour period. Ritonavir and cobicistat do not exclude in this arm (they exclude in the sildenafil arm, where the dose cannot be titrated within its cap); the 10mg in 72 hours limit applies.</b></p> <p>Mild to moderate hepatic impairment, Child-Pugh A or B. On-demand dose must not exceed 10mg.</p> <p>Moderate renal impairment, creatinine clearance 30 to 50 mL/min. For once-daily dosing start at 2.5mg.</p> <p>Severe renal impairment, creatinine clearance below 30 mL/min. On-demand dosing only; the dose must not exceed 10mg.</p> |
| <b>Actions if the patient is excluded or declines</b> | As for the sildenafil arm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## The medicine

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name, form and strength</b> | Tadalafil 2.5mg, 5mg, 10mg and 20mg film-coated tablets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Legal category</b>          | POM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Dose and frequency</b>      | <ul style="list-style-type: none"> <li>● <b>ON DEMAND: starting dose 10mg, at least 30 minutes before sexual activity. May be increased to 20mg or decreased to 5mg on efficacy and tolerability. MAXIMUM ONE DOSE IN ANY 24 HOURS.</b></li> <li>● ONCE DAILY: 2.5mg once daily at about the same time each day, increased to 5mg on tolerability. For men anticipating sexual activity at least twice per week.</li> <li>● <b>Do not use both regimens. Choose one and record which.</b></li> <li>● Effective for up to 36 hours. Absorption is not affected by food.</li> <li>● <b>With a potent CYP3A4 inhibitor, including ritonavir or cobicistat, on-demand use must not exceed 10mg in 72 hours.</b></li> <li>● Sexual stimulation is required.</li> </ul> |
| <b>Quantity to be supplied</b> | On demand: up to 8 tablets per supply. Once daily: up to 28 tablets, being one month. Decide the quantity on individual need and record the decision.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maximum or minimum treatment period</b> | No maximum treatment period. Review effectiveness, blood pressure and cardiovascular fitness at least annually, and reassess where there has been no improvement after 6 to 8 attempts at the maximum tolerated dose. For once-daily dosing, reassess periodically whether continued daily use remains appropriate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Route and method</b>                    | Oral.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Storage</b>                             | Store below 30 degrees Celsius in the original packaging. Protect from moisture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Adverse effects</b>                     | Very common: headache. Common: back pain, myalgia, pain in extremity, flushing, nasal congestion, dyspepsia, gastro-oesophageal reflux. Uncommon: dizziness, abdominal pain, hyperhidrosis, rash, tachycardia, palpitations. Rare but serious: priapism, non-arteritic anterior ischaemic optic neuropathy, sudden sensorineural hearing loss. Because of the prolonged half-life, adverse effects may persist longer than with sildenafil.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Yellow Card reporting</b>               | Report suspected adverse reactions via <a href="https://yellowcard.mhra.gov.uk">https://yellowcard.mhra.gov.uk</a> and inform the GP as appropriate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Records to be kept</b>                  | <ul style="list-style-type: none"> <li>• That valid informed consent was given.</li> <li>• Patient name, address, date of birth, and the GP with whom they are registered.</li> <li>• <b>THE ANSWER TO THE NITRATE AND NITRIC OXIDE DONOR QUESTIONS, including the direct question about poppers.</b></li> <li>• <b>THE CARDIOVASCULAR FITNESS ANSWER, in the patient's terms, and the blood pressure measured today.</b></li> <li>• <b>Whether the patient is on an alpha-blocker, and if so which one and whether he is stable on it.</b></li> <li>• Age, and where aged 65 or over, the starting dose chosen.</li> <li>• Any CYP3A4 inhibitor, and any hepatic or renal impairment, and the starting dose chosen as a result.</li> <li>• Name of the healthcare practitioner and registration number.</li> <li>• Name, brand, form, strength, dose, route, quantity and date of supply.</li> <li>• That the patient was advised these medicines do not protect against sexually transmitted infections.</li> <li>• Advice given, including advice given if the patient is excluded or declines.</li> <li>• Details of any adverse drug reactions and the actions taken.</li> <li>• That the medicine was supplied under this PGD.</li> </ul> <p>Records signed, dated, legible and contemporaneous. Electronic records are acceptable. Keep for 8 years.</p> |
| <b>Disposal</b>                            | Advise the patient to return unused tablets to a pharmacy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Information for the patient

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|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Written information</b> | Supply the patient information leaflet provided with the product.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Counselling</b>         | <ul style="list-style-type: none"> <li>• On demand: take at least 30 minutes before sex. It can still work up to 36 hours later.</li> <li>• Once daily: take at the same time each day, whether or not you expect to have sex.</li> <li>• <b>You still need to be aroused. The tablet does not do it on its own.</b></li> <li>• <b>One dose in 24 hours. No more.</b></li> <li>• <b>NEVER take this with poppers, or with any nitrate medicine such as a GTN spray or nicorandil. Because this medicine stays in your system for up to two days, that warning applies for two days after your last dose, not just on the day.</b></li> <li>• This does not protect you against sexually transmitted infections.</li> </ul> |

**Follow-up and urgent symptoms**

**Seek immediate medical attention for:**

- **An erection lasting more than 4 hours. Go to A&E.**
- **Sudden loss of vision in one or both eyes. Stop the tablets.**
- Sudden loss or reduction of hearing.
- **Chest pain during or after sex. Do not take a GTN spray for it; tell the paramedics you have taken this medicine, and that it lasts up to 36 hours.**

If you are not under regular GP review, arrange a cardiovascular and diabetes check.

## Appendix 1: Cardiovascular fitness assessment

Complete for every patient, at every supply, and record the answers. Sexual activity carries a cardiac workload comparable to brisk walking or climbing two flights of stairs.

|                                              |                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. The functional question</b>            | <p>"Can you walk a mile on the flat in about 20 minutes, or climb two flights of stairs briskly, without chest pain and without having to stop for breath?"</p> <ul style="list-style-type: none"><li>• YES: proceed.</li><li>• NO, or does not know: DO NOT SUPPLY. Refer to the GP.</li></ul>              |
| <b>2. Symptoms on exertion or during sex</b> | <ul style="list-style-type: none"><li>• Chest pain, breathlessness or palpitations on exertion, or during previous sexual activity.</li><li>• Any YES: DO NOT SUPPLY. Refer.</li></ul>                                                                                                                       |
| <b>3. Blood pressure, measured today</b>     | <ul style="list-style-type: none"><li>• Below 90/50: exclude.</li><li>• Above 170/100: exclude.</li><li>• Record the reading. Do not accept a figure from memory.</li></ul>                                                                                                                                  |
| <b>4. Recent cardiac events</b>              | <ul style="list-style-type: none"><li>• Stroke or myocardial infarction in the last 6 months: exclude.</li><li>• Unstable angina, angina during sex, heart failure of NYHA class 2 or greater in the last 6 months, or uncontrolled arrhythmia: exclude.</li></ul>                                           |
| <b>5. The opportunity</b>                    | <ul style="list-style-type: none"><li>• Erectile dysfunction commonly precedes a cardiovascular event by 2 to 5 years, and is often the first sign of undiagnosed diabetes.</li><li>• Where the patient has had no recent check, recommend one and record that you did, whether or not you supply.</li></ul> |

## Appendix 2: Every nitric oxide donor, by name

Ask about all of these.

|                                                |                                                                                                                                                                                                                                                                                      |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prescribed, obvious</b>                     | <ul style="list-style-type: none"><li>• Glyceryl trinitrate: tablets, spray, patch, ointment, infusion.</li><li>• Isosorbide mononitrate.</li><li>• Isosorbide dinitrate.</li></ul>                                                                                                  |
| <b>Prescribed, easily missed</b>               | <ul style="list-style-type: none"><li>• <b>NICORANDIL. An angina medicine. No "nitrate" in the name.</b></li><li>• Sodium nitroprusside.</li><li>• Riociguat and other soluble guanylate cyclase stimulators.</li></ul>                                                              |
| <b>Not prescribed, will not be on any list</b> | <ul style="list-style-type: none"><li>• <b>AMYL NITRITE, butyl nitrite, isobutyl nitrite, alkyl nitrites. Sold as "poppers", "room odouriser" or "leather cleaner".</b></li><li>• <b>Ask directly, without judgement, and record the answer. He will not volunteer it.</b></li></ul> |

## Appendix 3: Key references

- NICE Clinical Knowledge Summary: Erectile dysfunction.
- Summary of Product Characteristics for sildenafil and for tadalafil, current versions.
- British National Formulary.
- NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions, <https://www.nice.org.uk/guidance/mpg2>

### Authorisation

|                   |                                |
|-------------------|--------------------------------|
| <b>Version</b>    | v008                           |
| <b>Supersedes</b> | Version 007, 11 September 2026 |

|                        |                   |
|------------------------|-------------------|
| <b>Valid from date</b> | 11 September 2026 |
| <b>Expiry date</b>     | 31 July 2027      |

|                                                                                                                                                                                                     |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Authorised on behalf of Get Real Health by the Medical Director and the Head Pharmacist. Their signatures for this version are on the signed authorisation page at the end of this document.</b> |  |
|                                                                                                                                                                                                     |  |

#### **Adoption by the employing organisation**

*To be completed by the adopting pharmacy. These signatories are the adopting organisation own signatories and must not be pre-filled by Get Real Health.*

|                                       |                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------|
| <b>Superintendent / clinical lead</b> | Name: .....<br>Job title and organisation: .....<br>Signature: .....<br>Date: ..... |
| <b>Professional group signatory</b>   | Name: .....<br>Job title and organisation: .....<br>Signature: .....<br>Date: ..... |

## Change history

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| Version | v003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Date    | 8 September 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Changes | <ul style="list-style-type: none"><li>• The practitioner Agreement to practise page, the premises block and the practitioner signature table are restored. Every version 001 document carried them; no document produced by the current generator did, through one omission in one function, which left 15 live documents with no page for an individual practitioner to sign. Raised as blocking by an adopting pharmacy: NICE MPG2 expects a record of the individuals authorised to work under a PGD. The standard governance requirements are restored at the same time: SPC and BNF familiarity, MHRA safety alerts, CPD and appraisal, indemnity, and capacity and consent, all of which were in version 001 and were lost in the rewrite. This version changes no clinical content.</li><li>• Full clinical review and reissue. Re-read before working to it.</li><li>• One nitrate and nitric oxide donor exclusion now applies to BOTH arms, naming glyceryl trinitrate in all its forms, isosorbide mono- and dinitrate, nicorandil, amyl nitrite and the other alkyl nitrites, sodium nitroprusside and riociguat. The tadalafil arm previously said only "organic nitrates in any form", so amyl nitrite was invisible in it. Nicorandil appeared in neither arm.</li><li>• A required, recorded direct question about poppers, with suggested wording, because the patient will not volunteer it and it does not appear on any medication list.</li><li>• The cardiovascular fitness criterion has content: a functional question with a stated threshold, symptoms on exertion or during sex, a blood pressure measured on the day, and a requirement to record all of it. Appendix 1.</li><li>• Alpha-blocker starting dose restored to the tadalafil arm: 10mg on demand, 2.5mg once daily. The sildenafil arm keeps its 25mg rule.</li><li>• Over-65 dose reduction added to the tadalafil arm, which had none. Sildenafil 25mg, tadalafil 10mg on demand or 2.5mg once daily.</li><li>• New exclusions in both arms: previous priapism on a PDE5 inhibitor; concurrent use of another PDE5 inhibitor including one obtained online; and erectile dysfunction of sudden onset after trauma, surgery or a new medicine, or with penile pain or deformity, which needs a diagnosis rather than a tablet.</li><li>• Counselling now states that the tadalafil nitrate warning applies for up to two days after a dose because of its 17.5 hour half-life, and that a patient with chest pain after either medicine must tell the paramedics what he has taken rather than use a GTN spray.</li><li>• Records must carry the nitrate and poppers answers, the cardiovascular fitness answer and today's blood pressure, alpha-blocker status, and the reason for any reduced starting dose.</li><li>• Broken NICE reference corrected from mg2 to mpg2.</li></ul> |

## Previous versions

|     |                                                                                           |
|-----|-------------------------------------------------------------------------------------------|
| 001 | Development and issue of new PGD. Sildenafil and tadalafil arms, adult males 18 and over. |
|-----|-------------------------------------------------------------------------------------------|

**Standard requirements for every practitioner working under this PGD**

All users must be familiar with the current Summary of Product Characteristics for every product named in this PGD, and with current BNF and national guidance for the condition.

All users must be up to date with MHRA drug safety alerts and recalls relevant to these products.

All users must be up to date with their CPD requirements and appraisal.

All users must hold appropriate professional indemnity covering this service.

All users must understand capacity and consent, including the Mental Capacity Act 2005, and must be able to assess consent in children and young people where this PGD covers them.

### Agreement to practise by the registered healthcare professional

By signing below you confirm that you agree to the contents of this PGD and that you will work within it. A PGD does not remove the inherent professional obligations or accountability of the individual practitioner. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

**Name and address of the pharmacy or clinic premises to which this PGD relates**

Premises name .....  
 Address .....  
 ..... Postcode: .....  
 ODS code, if applicable .....

**Registered healthcare professionals authorised to work under this PGD at those premises**

The employing organisation must keep this page, or an equivalent local register, and must be able to produce it on request. A practitioner who has not signed against the current version is not authorised to work under it.

[illegible]

## **Version and change record**

Version: v008

Issued: 11 September 2026

Supersedes: Version 007, 11 September 2026

Valid from: 11 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Decision 45: the tadalafil caution and dose row now name ritonavir and cobicistat among the potent CYP3A4 inhibitors that cap on-demand use at 10mg in any 72 hour period, and state that these two medicines exclude in the sildenafil arm but not in the tadalafil arm; the arms are otherwise unchanged.

Decisions of the authorising signatories, 11 September 2026: the questions raised by the reviews of 10 and 11 September were put to the Medical Director and answered; each answer is written into this document above and, where the answer set a rule, into the consultation tool. Text drafted from a source for the Head Pharmacist to confirm is marked as such in the change lines.

## **Previous version records**

Version: v007

Issued: 11 September 2026

Supersedes: Version 006, 11 September 2026

Valid from: 11 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Tadalafil arm now carries the same 6 to 8 attempts reassessment rule as the sildenafil arm.

Age-related starting dose applies at 65 or over in both arms and in the records, which is the SmPC and BSSM basis; the previous wording was over 65.

Tadalafil severe renal impairment: the exclusion is the once-daily regimen; the 10mg on-demand cap now sits in the cautions where a dose cap belongs.

Appendix 1 question 4 now uses the exclusion criterion it summarises: heart failure of NYHA class 2 or greater in the last 6 months, not severe heart failure.

Wording corrections from the tool alignment and adversarial review of the consultation tools, 11 September 2026. Each correction resolves a contradiction inside the document or a plain error, taking the safer reading; no indication is widened, no dose raised and no exclusion removed. Questions that need a clinical decision are recorded separately and are not changed here.

Previous version records

Version: v006

Issued: 11 September 2026

Supersedes: Version 005, 10 September 2026

Valid from: 11 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Structural clean-up applied to every live Get Real Health PGD on 11 September 2026, following the adversarial review of 10 September: the adopting pharmacy's authorisation block is blank, for the pharmacy's own signatories; one signed authorisation page for this version replaces the earlier signature blocks; every validity cell states this version's dates (valid from 11 September 2026, expiry 31 July 2027); narration about earlier versions, the consultation tool and individual customers is removed from the body of the document; the change records of earlier reissues are carried below as previous version records. No clinical criterion changes unless listed.

Previous version records

Version: v004

Issued: 10 September 2026

Supersedes: Version 003, 9 September 2026

Change: the summary of the governing guidance is added as part 2 of the document, between the cover and the PGD, which is the house format for every Get Real Health PGD. It is summarised from the source it names and adds nothing to it. Nothing else in this document is altered, and this is not a clinical review of the remainder.

Version: v005

Issued: 10 September 2026

Supersedes: Version 004, 10 September 2026

Valid from: 10 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Validity: this version is valid from 10 September 2026 and expires on 31 July 2027. The signed master previously stated no period of validity, or an expiry of 31 October 2026; those are superseded.

Exclusion criteria, both arms: heart failure NYHA class 2 or greater in the last 6 months (Cialis SmPC 4.3), hypertrophic cardiomyopathy, significant aortic stenosis or other moderate to severe valve disease, and a murmur of unknown cause (BSSM high-risk group, sildenafil SmPC 4.4) added. The previous text excluded 'severe heart failure' only.

Sildenafil: ritonavir and cobicistat are exclusions. The SmPC caps sildenafil at 25 mg in 48 hours with ritonavir; the previous version allowed titration to 100 mg.

Tadalafil: doxazosin is an exclusion (SmPC: combination not recommended); severe renal impairment now caps on-demand dosing at 10 mg as well as excluding once-daily dosing.

## Signed on behalf of Get Real Health

Version v008, 11 September 2026.

| Name             | Qualification                    | Signature                                                                          | Date              |
|------------------|----------------------------------|------------------------------------------------------------------------------------|-------------------|
| Nitin Shori      | Medical Director<br>GMC: 6047293 |  | 11 September 2026 |
| Chris Pilkington | Head Pharmacist<br>GPhC: 2046322 |  | 11 September 2026 |

*Both authorising signatories reviewed this version together on 11 September 2026 and gave their authorisation for it to be issued. Signatures were applied digitally on their joint instruction, which is the established process for Get Real Health PGDs. This version is not valid without both signatures.*